THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If the 4th term of an AP is 9 and the 9th term is 4, then the 13th term is: 1

(a) 0 **(b)** 1 **(c)** 36 **(d)** 13

2. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 2$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 1: 1

(a) 10 **(b)** 4 **(c)** 9 **(d)** 8

3. The point on the y-axis equidistant from $(2, 3)$ and $(6, 5)$ has $y =$ 1

(a) 12 **(b)** -12 **(c)** 4 **(d)** 24

4. The sum of the first 13 terms of an AP is 390, and its first term is 6. The 13th term is: 1

(a) 66 **(b)** 54 **(c)** 24 **(d)** 60

5. In a right triangle with hypotenuse $c = 5$ and one leg $a = 3$, the area (sq. units) is: 1

(a) 10 **(b)** $\frac{15}{2}$ **(c)** 12 **(d)** 6

6. The area of the triangle formed by $(7, 0)$, $(0, 9)$ and the origin, doubled, is: 1

(a) $\frac{63}{2}$ **(b)** 63 **(c)** 16 **(d)** 126

7. In an AP, $S_8 = 200$ and $S_7 = 154$. Then $a_8 =$ 1

(a) 354 **(b)** 40 **(c)** 52 **(d)** 46

8. The altitude on the hypotenuse of the right triangle with legs 3 and 4 is of length (cm): 1

(a) 6 **(b)** $\frac{12}{5}$ **(c)** $\frac{5}{2}$ **(d)** $\frac{7}{2}$

9. If $P(x, y)$ is equidistant from $(1, 3)$ and $(7, 5)$ and lies on $y = 0$, then $x =$ 1

(a) $\frac{32}{3}$ **(b)** $-\frac{16}{3}$ **(c)** 4 **(d)** $\frac{16}{3}$

10. The sum of the first n terms of an AP is $S_n = 2n^2 + 3n$. Its common difference is: 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** 4

11. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 18$ and $ar(PQR) = 98$ and $AB = 6$, then $PQ =$ 1

(a) 98 **(b)** 6 **(c)** 17 **(d)** 14

12. The ratio in which $(-3, 2)$ and $(6, 5)$ are joined by the y-axis is 3: 1

(a) 5 **(b)** 6 **(c)** 9 **(d)** 3

13. **Assertion (A):** The list $3, 7, 11, 15, \ldots$ forms an AP. 1
Reason (R): In an AP the difference between any two consecutive terms is the same constant.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, then $EC = 6$ cm. 1
Reason (R): A line parallel to one side of a triangle, meeting the other two sides at distinct points, divides those sides in the same ratio.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The distance of the point $(3, 4)$ from the origin is 5 units. 1
Reason (R): The distance of a point (x, y) from the origin is $\sqrt{x^2 + y^2}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The 4th term of an AP is 9 and the 9th term is 4. Find the 13th term. 2

17. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $AB = 7$ cm and $BC = 14$ cm. Find DE. 2

18. Show that the points $A(0, 0)$, $B(3, 4)$, $C(8, 4)$, $D(5, 0)$ are the vertices of a rhombus. 2

19. If $S_n = 3n^2 + 4n$, find the first term and the common difference. 2

20. The areas of two similar triangles are 27 cm² and 75 cm². If a side of the first is 3 cm, find the corresponding side of the second. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In an AP, $S_6 = 75$ and $S_{10} = 185$. Find the AP. 3

22. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 8$ cm and $DC = 12$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

23. Show that the points $A(1, 1)$, $B(5, 4)$, $C(10, 4)$, $D(6, 1)$ are the vertices of a rhombus. 3

24. Find the sum of the first 16 terms of the AP whose 3rd term is 14 and 8th term is 34. 3

OR

Find the altitude of an equilateral triangle of side 6 cm, in terms of $\sqrt{3}$. 3

25. The line joining $(0,\ 0)$ and $(9,\ 6)$ is trisected by points P and Q. Find their coordinates. 3

OR

Find the sum of all integers between 10 and 200 that leave remainder 3 when divided by 7. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Mohan starts a monthly savings plan. He saves ₹ 100 in the first month and increases his saving by ₹ 20 every month. 5

(i) Write the first term and the common difference of the AP formed by his monthly savings.

(ii) How much does he save in the 10th month?

(iii) Find his total savings in the first 12 months.

(iv) In which month will he save ₹ 400, and what will his total savings be up to that month?

27. The vertices of $\triangle ABC$ are $A(0,\ 0)$, $B(12,\ 0)$ and $C(6,\ 8)$. Find the coordinates of the centroid and the area of the triangle. 5

OR

A loan is repaid in 18 monthly instalments; the first is ₹2500 and each later instalment is ₹150 more than the previous. Find the instalment in the last month and the total amount repaid. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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